TORSION AND VANISHING PROPERTIES OF THE COHOMOLOGY OF THE EXTENDED ALTERNATING CECH COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Extended alternating Cech complex of a sequence and its shape in each degree. recalled in the introduction of the same block, source lines 6908-7064. Explains why the complex vanishes outside the range of degrees.
- Base change of the extended alternating Cech complex: localising the complex at an element is the complex of the localised module over the localised ring. the statement counted by the sibling card S1-algebra-extended-alternating-Cech-is-colimit-koszul at 7099-7197 and the surrounding block; the author cites Lemma lemma-extended-alternating-base-change at proof lines 7086-7090. Identifies the localised cohomology with the cohomology of the localised complex.
- The extended alternating Cech complex is homotopy equivalent to zero when one of the elements is a unit. the statement counted by the sibling card S1-algebra-extended-alternating-homotopy-zero at 6980-7063, cited at proof lines 7090-7092. Supplies the vanishing that the annihilation statement reduces to.

Source reference. Original lemma 0G6K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20200–20200. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0G6I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20198–20198. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0G6J: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20199–20199. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let $f_1, \ldots, f_r \in R$. Let M be an R-module. Let H^q be the qth cohomology module of the extended alternation Čech complex of M. Then*

(1) *$H^q = 0$ if $q \notin [0, r]$,*
(2) *for $x \in H^i$ there exists an $n \geq 1$ such that $f_i^n x = 0$ for $i = 1, \ldots, r$,*
(3) *the support of H^q is contained in $V(f_1, \ldots, f_r)$,*
(4) *if there is an $f \in (f_1, \ldots, f_r)$ which acts invertibly on M, then $H^q = 0$.*

Proof. Part (1) follows from the fact that the extended alternating Čech complex is zero in degrees < 0 and $> r$. To prove (2) it suffices to show that for each i there exists an $n \geq 1$ such that $f_i^n x = 0$. To see this it suffices to show that $(H^q)_{f_i} = 0$. Since localization is exact, $(H^q)_{f_i}$ is the qth cohomology module of the localization of the extended alternating complex of M at f_i. By Lemma 0G6I this localization is the extended alternating Čech complex of M_{f_i} over R_{f_i} with respect to the images of $f_1, \ldots, f_r$ in R_{f_i}. Thus we reduce to showing that H^q is zero if f_i is invertible, which follows from Lemma 0G6J. Part (3) follows from the observation that $(H^q)_{f_i} = 0$ for all i that we just proved. To see part (4) note that in this case f acts invertibly on H^q and H^q is supported on $V(f)$ by (3). This forces H^q to be zero (small detail omitted). □